

Code :

```
#include <iostream>

#include <fstream>

#include <iomanip>

using namespace std;


// Base class

class Product

{

protected:

    string name;

    float price;


public:

    Product(string n, float p)

    {

        name = n;

        price = p;

    }


    string getName()

    {

        return name;

    }


    float getPrice()

    {
```

```
        return price;
    }
};

// Beverage class
class Beverage : public Product
{
public:
    Beverage(string n, float p) : Product(n, p)
    {
    }
};

// Snack class
class Snack : public Product
{
public:
    Snack(string n, float p) : Product(n, p)
    {
    }
};

// Dessert class
class Dessert : public Product
{
public:
    Dessert(string n, float p) : Product(n, p)
    {
    }
};
```

```

    }
};

int main()
{
    // Beverage items
    Beverage b1("Masala Chai", 40);
    Beverage b2("Nimbu Pani", 35);

    // Snack items
    Snack s1("Samosa", 25);
    Snack s2("Kachori", 30);

    // Dessert items
    Dessert d1("Rasgulla", 40);
    Dessert d2("Shrikhand", 60);

    // Store menu in file
    ofstream menuFile("menu.txt");

    menuFile << "===== CAFE MENU =====\n";

    menuFile << "\nBEVERAGES\n";
    menuFile << "1. Masala Chai - Rs. 40\n";
    menuFile << "2. Nimbu Pani - Rs. 35\n";

    menuFile << "\nSNACKS\n";

```

```
menuFile << "3. Samosa - Rs. 25\n";  
menuFile << "4. Kachori - Rs. 30\n";
```

```
menuFile << "\nDESSERTS\n";  
menuFile << "5. Rasgulla - Rs. 40\n";  
menuFile << "6. Shrikhand - Rs. 60\n";
```

```
menuFile.close();
```

```
// Read menu from file
```

```
ifstream readMenu("menu.txt");
```

```
string line;
```

```
cout << "\n";
```

```
while (getline(readMenu, line))  
{  
    cout << line << endl;  
}
```

```
readMenu.close();
```

```
// Arrays to store selected items
```

```
string itemName[20];
```

```
int itemQty[20];
```

```
float itemPrice[20];
```

```
int count = 0;
```

```
int choice;
```

```
int quantity;
```

```
char more;
```

```
float subtotal = 0;
```

```
// Taking order
```

```
do
```

```
{
```

```
    cout << "\nEnter item number: ";
```

```
    cin >> choice;
```

```
    cout << "Enter quantity: ";
```

```
    cin >> quantity;
```

```
    string name;
```

```
    float price;
```

```
    switch(choice)
```

```
    {
```

```
        case 1:
```

```
            name = b1.getName();
```

```
            price = b1.getPrice();
```

break;

case 2:

name = b2.getName();

price = b2.getPrice();

break;

case 3:

name = s1.getName();

price = s1.getPrice();

break;

case 4:

name = s2.getName();

price = s2.getPrice();

break;

case 5:

name = d1.getName();

price = d1.getPrice();

break;

case 6:

name = d2.getName();

price = d2.getPrice();

break;

default:

```
        cout << "Wrong choice!";  
        continue;  
    }
```

```
    itemName[count] = name;  
    itemQty[count] = quantity;  
    itemPrice[count] = price * quantity;
```

```
    subtotal = subtotal + itemPrice[count];
```

```
    count++;
```

```
    cout << "Do you want to add another item? (y/n): ";  
    cin >> more;
```

```
} while(more == 'y' || more == 'Y');
```

```
// Calculate GST and total
```

```
float gst = subtotal * 0.05;
```

```
float total = subtotal + gst;
```

```
// Display final bill
```

```
cout << "\n";
```

```
cout << "----- BILL ----- \n";
```

```
cout << left << setw(22) << "Item"
```

```
    << setw(8) << "Qty"
```

```
    << "Price" << endl;
```

```
cout << "-----\n";
```

```
for(int i = 0; i < count; i++)
```

```
{
```

```
    cout << left << setw(22) << itemName[i]
```

```
        << setw(8) << itemQty[i]
```

```
        << itemPrice[i] << endl;
```

```
}
```

```
cout << "-----\n";
```

```
cout << left << setw(22) << "Subtotal"
```

```
    << subtotal << endl;
```

```
cout << left << setw(22) << "GST (5%)"
```

```
    << gst << endl;
```

```
cout << "-----\n";
```

```
cout << left << setw(22) << "Total"
```

```
    << total << endl;
```

```
cout << "-----\n";
```

```
cout << "Thank You! Visit Again.\n";
```

```
// Store final bill in file
```

```
ofstream billFile("bill.txt");
```

```
billFile << "----- BILL ----- \n";
```

```
billFile << left << setw(22) << "Item"
```

```
    << setw(8) << "Qty"
```

```
    << "Price" << endl;
```

```
billFile << "-----\n";
```

```
for(int i = 0; i < count; i++)
```

```
{
```

```
    billFile << left << setw(22) << itemName[i]
```

```
        << setw(8) << itemQty[i]
```

```
        << itemPrice[i] << endl;
```

```
}
```

```
billFile << "-----\n";
```

```
billFile << left << setw(22) << "Subtotal"
```

```
    << subtotal << endl;
```

```
billFile << left << setw(22) << "GST (5%)"
    << gst << endl;

billFile << "-----\n";

billFile << left << setw(22) << "Total"
    << total << endl;

billFile << "-----\n";

billFile << "Thank You! Visit Again.\n";

billFile.close();

return 0;
}
```